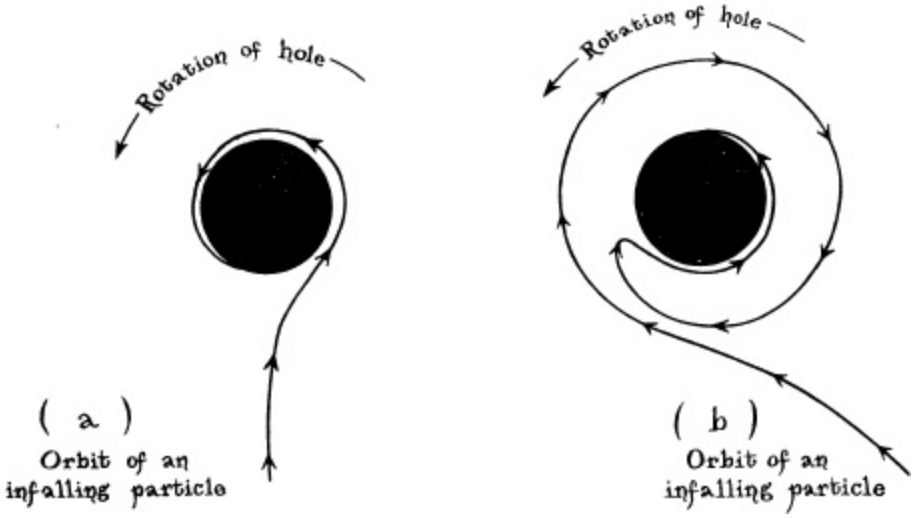
would move radially inward faster and faster at first; but then, as observed by the static, external observer, it would slow its infall and become frozen right at the horizon. (Recall the "frozen stars" of Chapter 6.) The hole's spin changes this in a very simple way: The spin makes space swirl, and the swirl of space makes the particle, as it nears the horizon, rotate in lockstep with the horizon itself. The particle thereby becomes frozen onto the spinning horizon and, as seen by the static, external observer, it circles around and around with the horizon forever. (Similarly, when a spinning star implodes to form a spinning hole, as seen by a static, external observer the star's surface "freezes" onto the spinning horizon, circling around and around with it forever.)



7.8 The trajectories in space of two particles that are thrown toward a black hole. (The trajectories are those that would be measured in a static, external reference frame.) Despite their very different initial motions, both particles are dragged, by the swirl of space, into precisely the same lockstep rotation with the hole as they near the horizon.

Though external observers see the particle of Figure 7.Ba freeze onto the spinning horizon and stay there forever, the particle itself sees something quite different. As the particle nears the horizon, gravitational time dilation forces the particle's time to flow more and more slowly, compared with the time of a static, external reference frame. When an infinite amount of external time has passed, the particle has experienced only a finite and very small amount of time. In that finite time, the particle has reached the hole's horizon, and in the next few moments of its time, it plunges right on through the horizon and down toward the hole's center. This enormous difference between the particle's infall as seen by the